

- 2 The continuous random variable X has cumulative distribution function F given by

$$F(x) = \begin{cases} 0 & x < -1, \\ \frac{1}{2}(1+x)^2 & -1 \leq x \leq 0, \\ 1 - \frac{1}{2}(1-x)^2 & 0 < x \leq 1, \\ 1 & x > 1. \end{cases}$$

- (a) Find the probability density function of X . [2]

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- (b) Find $P\left(-\frac{1}{2} \leq X \leq \frac{1}{2}\right)$. [2]

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(c) Find $E(X^2)$. [2]

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(d) Find $\text{Var}(X^2)$. [2]

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